

3. Project Design Phase

ComicCraft - AI/ML & Generative AI Project

1. System Design Overview

ComicCraft is designed as a web-based Generative AI application. The user enters a creative comic idea through the frontend. The request is sent to a FastAPI backend, which processes the prompt and uses Gemini AI to generate the comic story. The generated result is returned to the frontend and displayed as a structured four-panel comic story.

2. High-Level Architecture

User -> Web Interface -> FastAPI Backend -> Gemini AI -> Generated Story / Four Panels -> Web Interface

3. Main Modules

- User Interface Module - accepts the user's comic prompt and displays the generated result.
- API Module - receives frontend requests and returns generation responses.
- AI Generation Module - sends the prompt to Gemini AI and receives generated content.
- Response Processing Module - organizes the generated content for presentation.
- Configuration Module - loads API credentials and settings from environment variables.

4. Component Design

Component	Responsibility
Frontend	Prompt input, user interaction and result display.
FastAPI	Routing, request handling and backend orchestration.
Gemini AI	Generates the comic story and panel content from the prompt.
Environment Config	Stores configuration such as API credentials outside source code.

5. User Flow

- User opens the ComicCraft web application.
- User enters a creative story or comic idea.
- Frontend sends the prompt to the backend.
- FastAPI validates and processes the request.
- Gemini AI generates the requested comic content.
- The backend returns the generated response.
- Frontend displays the story in a four-panel structure.

6. Design Goals

- Simple and beginner-friendly interface
- Clear frontend-backend separation

- Reusable and maintainable architecture
- Secure handling of API credentials
- Easy future integration of image generation and comic export

7. Future Design Extension

The architecture can be extended to generate an image for each panel, add character consistency, speech bubbles, visual style selection, and PDF/image export without changing the core user-to-AI workflow.